JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Functions of Two or Three Real Variables (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
42	66 M	126 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	23	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	13	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q34 • ID: JAM_2021_Q34) **MSQ** +2 M Neg: Nil (0)

JAM 2021 MATHEMATICS - MA

Q. 34 Consider the two functions $f(x, y) = x + y$ and $g(x, y) = xy - 16$ defined on $\mathbb{R}^2$. Then

(A) the function f has no global extreme value subject to the condition $g = 0$.

(B) the function f attains global extreme values at $(4, 4)$ and $(-4, -4)$ subject to the condition $g = 0$.

(C) the function g has no global extreme value subject to the condition $f = 0$.

(D) the function g has a global extreme value at $(0, 0)$ subject to the condition $f = 0$.

Question 2 (JAM 2021 • Q33 • ID: JAM_2021_Q33) **MSQ** +2 M Neg: Nil (0)

Q. 33 Let $D = \mathbb{R}^2 \setminus \{(0, 0)\}$. Consider the two functions $u, v : D \to \mathbb{R}$ defined by

$$u(x, y) = x^2 - y^2 \text{ and } v(x, y) = xy.$$

Consider the gradients ∇u and ∇v of the functions u and v, respectively. Then

(A) ∇u and ∇v are parallel at each point (x, y) of D.

(B) ∇u and ∇v are perpendicular at each point (x, y) of D.

(C) ∇u and ∇v do not exist at some points (x, y) of D.

(D) ∇u and ∇v at each point (x, y) of D span $\mathbb{R}^2$.

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Question 3 (JAM 2021 • Q23 • ID: JAM_2021_Q23) **MCQ** **+2 M** **Neg: −0.67**

Q. 23 Let $D \subseteq \mathbb{R}^2$ be defined by $D = \mathbb{R}^2 \setminus \{(x, 0) : x \in \mathbb{R}\}$. Consider the function $f : D \to \mathbb{R}$ defined by

$$f(x, y) = x \sin \frac{1}{y}.$$

Then

(A) f is a discontinuous function on D.

(B) f is a continuous function on D and cannot be extended continuously to any point outside D.

(C) f is a continuous function on D and can be extended continuously to $D \cup \{(0, 0)\}$.

(D) f is a continuous function on D and can be extended continuously to the whole of $\mathbb{R}^2$.

Question 4 (JAM 2020 • Q53 • ID: JAM_2020_Q53) **NAT** **+2 M** **Neg: Nil (0)**

Q. 53 The minimum value of the function $f(x, y) = x^2 + xy + y^2 - 3x - 6y + 11$ is ________.

Question 5 (JAM 2020 • Q45 • ID: JAM_2020_Q45) **NAT** **+1 M** **Neg: Nil (0)**

Q. 45 Let $f(x, y) = e^x \sin y$, $x = t^3 + 1$ and $y = t^4 + t$. Then $\frac{df}{dt}$ at $t = 0$ is ________. (rounded off to two decimal places)

Question 6 (JAM 2020 • Q17 • ID: JAM_2020_Q17) MCQ +2 M Neg: −0.67

Q. 17 Let $f(x, y, z) = x^3 + y^3 + z^3 - 3xyz$. A point at which the gradient of the function f is equal to zero is

(A) $(-1, 1, -1)$ (B) $(-1, -1, -1)$ (C) $(-1, 1, 1)$ (D) $(1, -1, 1)$

Question 7 (JAM 2020 • Q15 • ID: JAM_2020_Q15) MCQ +2 M Neg: −0.67

Q. 15 Let $f(x,y) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y}, & y \neq 0, x = 0 \\ 0, & x = y = 0. \end{cases}$

Which of the following is true at $(0, 0)$?

(A) f is not continuous

(B) $\dfrac{\partial f}{\partial x}$ is continuous but $\dfrac{\partial f}{\partial y}$ is not continuous

(C) f is not differentiable

(D) f is differentiable but both $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ are not continuous

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Question 8 (JAM 2020 • Q6 • ID: JAM_2020_Q6) MCQ +1 M Neg: −0.33

Q. 6 If the directional derivative of the function $z = y^2 e^{2x}$ at $(2, -1)$ along the unit vector $\vec{b} = \alpha\hat{i} + \beta\hat{j}$ is zero, then $|\alpha + \beta|$ equals

(A) $\dfrac{1}{2\sqrt{2}}$ (B) $\dfrac{1}{\sqrt{2}}$ (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Question 9 (JAM 2020 • Q5 • ID: JAM_2020_Q5) **MCQ** **+1 M** **Neg: −0.33**

JAM 2020 MATHEMATICS - MA

Q. 5 If the equation of the tangent plane to the surface $z = 16 - x^2 - y^2$ at the point $P(1, 3, 6)$ is $ax + by + cz + d = 0$, then the value of $|d|$ is

(A) 16 (B) 26 (C) 36 (D) 46

Question 10 (JAM 2020 • Q4 • ID: JAM_2020_Q4) **MCQ** **+1 M** **Neg: −0.33**

Q. 4 Let $g : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function. If $f(x, y) = g(y) + xg'(y)$, then

(A) $\dfrac{\partial f}{\partial x} + y\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial f}{\partial y}$ (B) $\dfrac{\partial f}{\partial y} + y\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial f}{\partial x}$

(C) $\dfrac{\partial f}{\partial x} + x\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial f}{\partial y}$ (D) $\dfrac{\partial f}{\partial y} + x\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial f}{\partial x}$

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Question 11 (JAM 2019 • Q54 • ID: JAM_2019_Q54) **NAT** **+2 M** **Neg: Nil (0)**

Q.54 The number of critical points of the function

$$f(x, y) = (x^2 + 3y^2)e^{-(x^2+y^2)}$$

is ___________

Question 12 (JAM 2019 • Q48 • ID: JAM_2019_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let

$$f(x,y)=\begin{cases}\dfrac{x^3+y^3}{x^2-y^2}, & x^2-y^2\neq 0\\ 0, & x^2-y^2=0.\end{cases}$$

Then the directional derivative of f at $(0,0)$ in the direction of $\frac{4}{5}\hat{\imath}+\frac{3}{5}\hat{\jmath}$ is __________

Question 13 (JAM 2019 • Q39 • ID: JAM_2019_Q39) MSQ +2 M Neg: Nil (0)

Q.39 Let

$$f(x,y)=\begin{cases}\dfrac{|x|}{|x|+|y|}\sqrt{x^4+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0).\end{cases}$$

Then at $(0,0)$,

(A) f is continuous
(B) $\frac{\partial f}{\partial x}=0$ and $\frac{\partial f}{\partial y}$ does not exist
(C) $\frac{\partial f}{\partial x}$ does not exist and $\frac{\partial f}{\partial y}=0$
(D) $\frac{\partial f}{\partial x}=0$ and $\frac{\partial f}{\partial y}=0$

Question 14 (JAM 2019 • Q29 • ID: JAM_2019_Q29) MCQ +2 M Neg: −0.67

Q.29 For $\beta\in\mathbb{R}$, define

$$f(x,y)=\begin{cases}\dfrac{x^2|x|^\beta y}{x^4+y^2}, & x\neq 0,\\ 0, & x=0.\end{cases}$$

Then, at $(0,0)$, the function f is

(A) continuous for $\beta=0$
(B) continuous for $\beta>0$
(C) not differentiable for any β
(D) continuous for $\beta<0$

Question 15 (JAM 2019 • Q26 • ID: JAM_2019_Q26) MCQ +2 M Neg: −0.67

Q.26 The tangent line to the curve of intersection of the surface $x^2 + y^2 - z = 0$ and the plane $x + z = 3$ at the point $(1, 1, 2)$ passes through

(A) $(-1, -2, 4)$ (B) $(-1, 4, 4)$ (C) $(3, 4, 4)$ (D) $(-1, 4, 0)$

Question 16 (JAM 2019 • Q22 • ID: JAM_2019_Q22) MCQ +2 M Neg: −0.67

Q.22 The function

$$f(x, y) = x^3 + 2xy + y^3$$

has a saddle point at

(A) $(0, 0)$ (B) $\left(-\frac{2}{3}, -\frac{2}{3}\right)$ (C) $\left(-\frac{3}{2}, -\frac{3}{2}\right)$ (D) $(-1, -1)$

Question 17 (JAM 2019 • Q8 • ID: JAM_2019_Q8) MCQ +1 M Neg: −0.33

Q.8 The equation of the tangent plane to the surface $x^2z + \sqrt{8 - x^2 - y^4} = 6$ at the point $(2, 0, 1)$ is

(A) $2x + z = 5$
(B) $3x + 4z = 10$
(C) $3x - z = 10$
(D) $7x - 4z = 10$

Question 18 (JAM 2019 • Q3 • ID: JAM_2019_Q3) MCQ +1 M Neg: −0.33

Q.3 Let $g: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function. Define $f: \mathbb{R}^3 \to \mathbb{R}$ by

$$f(x, y, z) = g(x^2 + y^2 - 2z^2).$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

(A) $4(x^2 + y^2 - 4z^2)\, g''(x^2 + y^2 - 2z^2)$
(B) $4(x^2 + y^2 + 4z^2)\, g''(x^2 + y^2 - 2z^2)$
(C) $4(x^2 + y^2 - 2z^2)\, g''(x^2 + y^2 - 2z^2)$
(D) $4(x^2 + y^2 + 4z^2)\, g''(x^2 + y^2 - 2z^2) + 8g'(x^2 + y^2 - 2z^2)$

Question 19 (JAM 2018 • Q53 • ID: JAM_2018_Q53) NAT +2 M Neg: Nil (0)

Q.53 Suppose x, y, z are positive real numbers such that $x + 2y + 3z = 1$. If M is the maximum value of xyz^2 , then the value of $\frac{1}{M}$ is ________________

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Question 20 (JAM 2018 • Q45 • ID: JAM_2018_Q45) NAT +1 M Neg: Nil (0)

Q.45 Let $f(x,y) = \sqrt{x^3 y}\ \sin\left(\frac{\pi}{2}\ e^{\left(\frac{y}{x}-1\right)}\right) + xy\ \cos\left(\frac{\pi}{3}\ e^{\left(\frac{x}{y}-1\right)}\right)$ for $(x,y) \in \mathbb{R}^2,\ x > 0,\ y > 0$.

Then $f_x(1,1) + f_y(1,1) =$ ________________

Question 21 (JAM 2018 • Q44 • ID: JAM_2018_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} \dfrac{x^2 y\,(x-y)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Then $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$ at the point $(0,0)$ is _____________

Question 22 (JAM 2018 • Q42 • ID: JAM_2018_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let $\phi(x,y,z) = 3y^2 + 3yz$ for $(x,y,z) \in \mathbb{R}^3$. Then the absolute value of the directional derivative of ϕ in the direction of the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z}{-2}$, at the point $(1,-2,1)$ is ______________

Question 23 (JAM 2018 • Q18 • ID: JAM_2018_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $a, b \in \mathbb{R}$ and let $f\colon \mathbb{R} \to \mathbb{R}$ be a thrice differentiable function. If $z = e^u f(v)$, where $u = ax + by$ and $v = ax - by$, then which one of the following is TRUE?

(A) $b^2 z_{xx} - a^2 z_{yy} = 4a^2 b^2 e^u f'(v)$

(B) $b^2 z_{xx} - a^2 z_{yy} = -4e^u f'(v)$

(C) $b z_x + a z_y = abz$

(D) $b z_x + a z_y = -abz$

Question 24 (JAM 2018 • Q17 • ID: JAM_2018_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $f(x,y) = \begin{cases} \frac{xy}{(x^2+y^2)^\alpha}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

Then which one of the following is TRUE for f at the point $(0,0)$?

(A) For $\alpha = 1$, f is continuous but not differentiable

(B) For $\alpha = \frac{1}{2}$, f is continuous and differentiable

(C) For $\alpha = \frac{1}{4}$, f is continuous and differentiable

(D) For $\alpha = \frac{3}{4}$, f is neither continuous nor differentiable

Question 25 (JAM 2018 • Q6 • ID: JAM_2018_Q6) MCQ +1 M Neg: −0.33

Q.6 In $\mathbb{R}^3$, the cosine of the acute angle between the surfaces $x^2 + y^2 + z^2 - 9 = 0$ and $z - x^2 - y^2 + 3 = 0$ at the point $(2,1,2)$ is

(A) $\frac{8}{5\sqrt{21}}$ (B) $\frac{10}{5\sqrt{21}}$ (C) $\frac{8}{3\sqrt{21}}$ (D) $\frac{10}{3\sqrt{21}}$

Question 26 (JAM 2018 • Q5 • ID: JAM_2018_Q5) MCQ +1 M Neg: −0.33

Q.5 The tangent plane to the surface $z = \sqrt{x^2 + 3y^2}$ at $(1, 1, 2)$ is given by

(A) $x - 3y + z = 0$
(B) $x + 3y - 2z = 0$
(C) $2x + 4y - 3z = 0$
(D) $3x - 7y + 2z = 0$

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Question 27 (JAM 2017 • Q44 • ID: JAM_2017_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let P be the point on the surface $z = \sqrt{x^2 + y^2}$ closest to the point $(4,2,0)$. Then the square of the distance between the origin and P is ___________

Question 28 (JAM 2017 • Q33 • ID: JAM_2017_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be a function. Then which of the following statements is/are TRUE?

(A) If f is differentiable at $(0,0)$, then all directional derivatives of f exist at $(0,0)$

(B) If all directional derivatives of f exist at $(0,0)$, then f is differentiable at $(0,0)$

(C) If all directional derivatives of f exist at $(0,0)$, then f is continuous at $(0,0)$

(D) If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous in a disc centered at $(0,0)$, then f is differentiable at $(0,0)$

Question 29 (JAM 2017 • Q23 • ID: JAM_2017_Q23) MCQ +2 M Neg: −0.67

Q.23 Let $f(x,y) = \frac{x^2y}{x^2+y^2}$ for $(x,y) \neq (0,0)$. Then

(A) $\frac{\partial f}{\partial x}$ and f are bounded

(B) $\frac{\partial f}{\partial x}$ is bounded and f is unbounded

(C) $\frac{\partial f}{\partial x}$ is unbounded and f is bounded

(D) $\frac{\partial f}{\partial x}$ and f are unbounded

Question 30 (JAM 2017 • Q5 • ID: JAM_2017_Q5) MCQ +1 M Neg: −0.33

Q.5 Let $f: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function. If $g(u,v) = f(u^2 - v^2)$, then

$$\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} =$$

(A) $4(u^2 - v^2)f''(u^2 - v^2)$

(B) $4(u^2 + v^2)f''(u^2 - v^2)$

(C) $2f'(u^2 - v^2) + 4(u^2 - v^2)f''(u^2 - v^2)$

(D) $2(u - v)^2 f''(u^2 - v^2)$

Question 31 (JAM 2017 • Q1 • ID: JAM_2017_Q1) MCQ +1 M Neg: −0.33

Q.1 Consider the function $f(x,y) = 5 - 4\sin x + y^2$ for $0 < x < 2\pi$ and $y \in \mathbb{R}$. The set of critical points of $f(x,y)$ consists of

(A) a point of local maximum and a point of local minimum
(B) a point of local maximum and a saddle point
(C) a point of local maximum, a point of local minimum and a saddle point
(D) a point of local minimum and a saddle point

Question 32 (JAM 2016 • Q60 • ID: JAM_2016_Q60) NAT +2 M Neg: Nil (0)

Q.60 The maximum value of $f(x,y) = x^2 + 2y^2$ subject to the constraint $y - x^2 + 1 = 0$ is ____________

END OF THE QUESTION PAPER

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Question 33 (JAM 2016 • Q44 • ID: JAM_2016_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let f be a real valued function defined by

$$f(x,y) = 2\ln\left(x^2y^2e^{\frac{y}{x}}\right), \qquad x > 0, y > 0.$$

Then the value of $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ at any point (x,y), where $x > 0$, $y > 0$, is ____________

Question 34 (JAM 2016 • Q33 • ID: JAM_2016_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{xy}{|x|} & \text{if } x \neq 0 \\ 0 & \text{elsewhere} \end{cases}.$$

Then at the point $(0,0)$, which of the following statement(s) is(are) TRUE ?

(A) f is not continuous
(B) f is continuous
(C) f is differentiable
(D) Both first order partial derivatives of f exist

Question 35 (JAM 2016 • Q18 • ID: JAM_2016_Q18) MCQ +2 M Neg: −0.67

Q.18 The function $f(x,y) = 3x^2y + 4y^3 - 3x^2 - 12y^2 + 1$ has a saddle point at

(A) $(0,\ 0)$ (B) $(0,\ 2)$ (C) $(1,\ 1)$ (D) $(-2,\ 1)$

Question 36 (JAM 2016 • Q17 • ID: JAM_2016_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{xy^2}{x+y} & \text{if } x+y \neq 0 \\ 0 & \text{if } x+y = 0 \end{cases}.$$

Then the value of $\left(\dfrac{\partial^2 f}{\partial x\, \partial y} + \dfrac{\partial^2 f}{\partial y\, \partial x}\right)$ at the point $(0,0)$ is

(A) 0 (B) 1 (C) 2 (D) 4

Question 37 (JAM 2015 • Q45 • ID: JAM_2015_Q45) NAT +1 M Neg: Nil (0)

Q.5 Let $f\colon \mathbb{R}^3 \longrightarrow \mathbb{R}$ be defined by

$$f(x,y,z) = \sin x + 2\, e^{\frac{y}{2}} + z^2$$

The maximum rate of change of f at $\left(\frac{\pi}{4},\ 0,\ 1\right)$, correct upto three decimal places, is

____

Question 38 (JAM 2015 • Q44 • ID: JAM_2015_Q44) NAT +1 M Neg: Nil (0)

Q.4

Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \left(1 + \frac{x}{y}\right)^2, & y \neq 0 \\ 0, & y = 0 \end{cases}$$

If the directional derivative of f at $(0, 0)$ exists along the direction $\cos\alpha\ \hat{\imath} + \sin\alpha\ \hat{\jmath}$, where $\sin\alpha \neq 0$, then the value of $\cot\alpha$ is ____________

Question 39 (JAM 2015 • Q38 • ID: JAM_2015_Q38) MSQ +2 M Neg: Nil (0)

Q.8 Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} xy\frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0,0) \end{cases}$$

At $(0, 0)$,

(A) f is not continuous

(B) f is continuous, and both f_x and f_y exist

(C) f is differentiable

(D) f_x and f_y exist but f is not differentiable

Question 40 (JAM 2015 • Q30 • ID: JAM_2015_Q30) MCQ +2 M Neg: −0.67

Q.30

Let $f : \{ (x, y) \in \mathbb{R}^2 : x > 0,\ y > 0 \} \to \mathbb{R}$ be given by

$$f(x, y) = x^{\frac{1}{3}}\, y^{\frac{-4}{3}} \tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{\sqrt{x^2 + y^2}}$$

Then the value of

$$g(x, y) = \frac{x f_x(x,y) + y f_y(x,y)}{f(x,y)}$$

(A) changes with x but not with y

(B) changes with y but not with x

(C) changes with x and also with y

(D) neither changes with x nor with y

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 1 – Q. 10 carry two marks each.

Question 41 (JAM 2015 • Q25 • ID: JAM_2015_Q25) MCQ +2 M Neg: −0.67

Q.25

For what real values of x and y, does the integral $\int_x^y (6 - t - t^2)dt$ attain its maximum?

(A) $x = -3,\ y = 2$ (B) $x = 2,\ y = 3$
(C) $x = -2,\ y = 2$ (D) $x = -3,\ y = 4$

Question 42 (JAM 2015 • Q15 • ID: JAM_2015_Q15) MCQ +2 M Neg: −0.67

Q.15 Suppose that the dependent variables z and w are functions of the independent variables x and y, defined by the equations $f(x, y, z, w) = 0$ and $g(x, y, z, w) = 0$, where $f_z g_w - f_w g_z = 1$. Which one of the following is correct?

(A) $z_x = f_w g_x - f_x g_w$
(B) $z_x = f_x g_w - f_w g_x$
(C) $z_x = f_z g_x - f_x g_z$
(D) $z_x = f_z g_w - f_z g_x$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of Two or Three Real Variables (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MSQ	+2	**A, D**
2	MSQ	+2	**B, D**
3	MCQ	+2	**C**
4	NAT	+2	**2 to 2**
5	NAT	+1	**2.70 to 2.72**
6	MCQ	+2	**B**
7	MCQ	+2	**D**
8	MCQ	+1	**C**
9	MCQ	+1	**Marks to All**
10	MCQ	+1	**C**
11	NAT	+2	**5 to 5**
12	NAT	+1	**2.6 to 2.6**
13	MSQ	+2	**A, D**
14	MCQ	+2	**B**

Q. No.	Type	Marks	Official Key
15	MCQ	+2	**B**
16	MCQ	+2	**A**
17	MCQ	+1	**B**
18	MCQ	+1	**B**
19	NAT	+2	**1140 to 1160**
20	NAT	+1	**3 to 3**
21	NAT	+1	**1 to 1**
22	NAT	+1	**6.5 to 7.5**
23	MCQ	+2	**A**
24	MCQ	+2	**C**
25	MCQ	+1	**C**
26	MCQ	+1	**B**
27	NAT	+1	**9.9 to 10.1**
28	MSQ	+2	**A, D**

Q. No.	Type	Marks	Official Key
29	MCQ	+2	**B**
30	MCQ	+1	**B**
31	MCQ	+1	**D**
32	NAT	+2	**2.0 to 2.0**
33	NAT	+1	**8.0 to 8.0**
34	MSQ	+2	**B;D**
35	MCQ	+2	**D**
36	MCQ	+2	**B**
37	NAT	+1	**2.34 to 2.35**
38	NAT	+1	**-1**
39	MSQ	+2	**B;C**
40	MCQ	+2	**D**
41	MCQ	+2	**A**
42	MCQ	+2	**A**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.